

Contents

Known bugs and v1.0 limitations	1
v1.0 known limitations (workarounds documented; v1.1 candidates)	1
.dta round-trip loses xtset state	2
ARIMA uses conditional likelihood, not exact ML (no Kalman filter)	2
Factor variable composition with TS ops	2
xtreg, be doesn't weight by T_i	2
MA(1) and ARMA SE precision	2
Variable-name length limit (32 chars)	2
import excel without firstrow still treats row 1 as headers	3
No graphics	3
CHANGELOG (recently fixed)	3
Recently fixed	3
Tier 4 — econometric completion (NEW for v1.0)	3
Tier 3 — output and reproducibility (NEW)	4
Tier 2 — factor variables: i., c., #, ##, ib. (NEW)	5
Tier 1 polish — statistical functions, egen extensions, _b/_se, formats	6
margins — marginal effects after regress / logit / probit (NEW)	7
predict extended to logit, probit, and xtreg (NEW)	7
logit and probit — binary-outcome MLE (NEW)	8
test syntax extended (NEW)	9
lincom accepts TS-op coefficient names (NEW)	9
xtreg, re — random-effects (GLS) estimator (NEW)	9
hausman — FE vs RE specification test (NEW)	9
xtreg, fe — fixed-effects (within) estimator (NEW)	10
TS-op handling unified across all varlist-accepting commands (NEW)	10
xtdescribe / xtides (NEW)	11
TS operators in summarize and list (FIXED)	11
D-operator iterated differencing (FIXED — silent math bug)	11
Operator-list TS syntax (FIXED)	11
Multi-digit time-series operators read wrong lag (FIXED)	12
Time-series operators rejected in regress varlist (FIXED)	12
capture and _rc (FIXED in v1.4.0)	12
test 08 captures fewer lines than expected	12
Storage types are display-only	13
Numeric function family coverage	13
v0.6 .dta polish (this milestone)	13
Still deferred from v0.6	13
reported by user during real-data testing	13
reported by Claude during synthetic testing	14
reported by Mico during WB fertility replication (v1.6.x)	14

Known bugs and v1.0 limitations

v1.0 known limitations (workarounds documented; v1.1 candidates)

These are not bugs in the strict sense — they're scope decisions for v1.0 that we may revisit based on real-world testing.

.dta round-trip loses xtset state

save mydata.dta then use mydata.dta preserves variable types, formats, variable labels, and value labels, but does **not** preserve the xtset panel/time variable assignment. Stata writes this as a dataset characteristic (_dta[iis], _dta[tis]); readstat doesn't expose characteristic writing in its public API, and hand-writing characteristic blocks in raw .dta format is a v1.1 task.

Workaround: re-run xtset country year after use. This is the standard Stata-do-file convention anyway, so most users won't notice.

ARIMA uses conditional likelihood, not exact ML (no Kalman filter)

tea's arima command computes parameters by minimizing the SSR of the recursively-computed residuals, conditioning on $\varepsilon_{\{<0\}}=0$ (the conditional likelihood approach). This gives consistent estimates but slightly different finite-sample behavior than Stata's default exact ML via the Kalman filter:

- Absolute log-likelihood values differ from Stata's (we drop the initial-state contribution)
- AR coefficients near the unit-root boundary may converge less robustly
- No seasonal ARIMA (sar, sma) in v1.0

Workaround: for serious time-series work, escape to R's arima() or Python's statsmodels.tsa.arima.

Factor variable composition with TS ops

i.country#c.L.gdp (factor interaction with a lagged regressor) is **not** supported directly. The factor module operates on column names, not on materialized TS-op temps.

Workaround: pre-compute the lag manually: gen L1_gdp = L.gdp, then use i.country#c.L1_gdp.

xtreg, be doesn't weight by T_i

Stata's default xtreg, be uses a slight variance correction for unbalanced panels (down-weights groups with few observations). tea uses straight OLS on the panel means. For balanced panels the answers are identical. For unbalanced panels they can differ by a few percent in the SEs.

MA(1) and ARMA SE precision

ARIMA standard errors are computed from a numerical-differentiation Hessian (forward differences scaled by 1e-5). For models with $q>0$ (MA components), the SEs can be off by 1-5% compared to analytical or exact-ML Hessians. Point estimates are unaffected.

Variable-name length limit (32 chars)

A residual Stata-12 limit. Factor-variable interaction names like 2010.year#1985.country#c.gdp (28 chars) fit; longer 3-way interactions truncate. In practice not a problem.

import excel without firstrow still treats row 1 as headers

Stata's rule when *firstrow* is *omitted*: name every column by its Excel letter (A, B, ...) and keep row 1 as a data row. *tea* currently always consumes row 1 as headers, whether or not *firstrow* was given — so omitting *firstrow* silently loses the first data row. With *firstrow* given, *tea* now matches Stata exactly (v1.6.x: invalid chars removed, digit-leading headers like 1960 fall back to the column letter). Fixing the no-*firstrow* case means restructuring *load_csv_into*'s header/data row accounting; it also changes behavior for any existing *tea* script that omitted *firstrow* and relied on the lenient reading. Deferred pending a scope decision.

Workaround: always pass *firstrow* (which Stata do-files written against WB/WEO-style sheets do anyway), or add a header row to the sheet.

No graphics

tea ships zero plotting commands. Users wanting plots should export delimited and pipe to *gnuplot*, R's *ggplot2*, or Python's *matplotlib/seaborn*.

CHANGELOG (recently fixed)

Recently fixed

Tier 4 — econometric completion (NEW for v1.0)

ivregress 2sls (in *src/regress.c* ~300 lines):

Syntax: `ivregress 2sls y [exog] (endog = instruments) [, vce(robust|cluster v)]`. Full first-stage regression of each endogenous regressor on the combined instrument matrix $W = [\text{exog}, Z, _cons]$ via *dgels*, second-stage $\beta = (\hat{X}'\hat{X})^{-1}\hat{X}'y$; residuals computed using ORIGINAL X (not fitted) for SE; sandwich $V = A \cdot B \cdot A$ for robust/cluster ($A = (\hat{X}'\hat{X})^{-1}$, B uses original- X residuals). HC1 / CR1 finite-sample adjustments. First-stage F-stat reported (minimum across endogenous regressors — the weak-instrument worry case).

Verified on textbook endogeneity case: with $\text{cov}(x, u) \neq 0$, OLS biased to 2.49; IV recovers 1.997 against true 2.0, first-stage $F=1364$.

poisson — count outcomes via the existing MLE driver:

Added `mle_family_poisson` to *src/mle.c* with `poisson_per_obs`: $\mu = \exp(\eta)$ clipped at $\eta < 700$; `score=y-μ`; `weight=μ`; `loglik=yη-μ` (drops the $\log(y!)$ constant — absolute loglik differs from Stata but β identical). Added `count_validate` (rejects $y < 0$). Generalized `do_glm_binary` header text to "Poisson regression". Verified intercept-only: $\beta_cons = \log(\bar{y}) = 0.6931$ for $\bar{y}=2$ (exact).

xtreg, be (between estimator):

Replaced the "not implemented" stub. After group-mean computation (already done in the FE/RE path), branch into BE: build $G_obs \times (K_orig+1)$ matrix Xb (panel means + appended 1s column for cons), OLS via *dgels*, $\sigma^2_be = \text{RSS}/(G-K)$, $V = \sigma^2 \cdot (\hat{X}'\hat{X})^{-1}$ via Cholesky. Verified hand-checkable case: panel data with $\bar{y}_i = 5 + 2\bar{x}_i$ recovers $\beta_x=2$, $_cons=5$, $R^2_b=1.0000$ exactly.

arma (new module *src/arma.c*, ~580 lines):

Conditional-likelihood ARIMA(p,d,q) with optional ARIMAX exog regressors. Syntax: `arima y [exog], arima(p d q) [noconstant]`. Algorithm: - Difference y in place d times - Pack params as $[\mu, \beta_K, \phi_p, \theta_q]$ - Recursive residual computation with $\varepsilon_{\{<0\}}=0$ conditional assumption - Gauss-Newton iteration with step halving (max 50 iterations) - SSR via finite-difference gradient + Hessian - $V \approx 2\sigma^2 \cdot \text{Hess}^{-1}$ via `dgetrf/dgetri` with ridge

Verified: AR(1) on strong-signal data recovers $\phi=0.7010$ (true 0.7). AR(1) on US WEO growth data matches OLS lag regression to 3 digits ($\hat{\phi} = 0.154$ both). MA(1) Monte Carlo with `rnormal()` noise: $\theta=0.429$ (true 0.5) at N=500.

`runiform()` and `rnormal()` (in `eval.c`):

PCG32-based PRNG with Box-Muller normal. `set seed N` for reproducibility. Essential for Monte Carlo, simulation, and bootstrap workflows. Tested against expected distributional moments: mean 0.492, sd 0.291 for u (vs true 0.5, 0.289); mean 0.001, sd 1.015 for z (vs true 0, 1).

Tier 3 — output and reproducibility (NEW)

estimates command suite (new module `src/estcmd.c`):

```
estimates store NAME      clone last_est, save under name
estimates restore NAME    load NAME → last_est (for predict/margins/test)
estimates dir             list saved with cmd, N, depvar
estimates drop NAME...    remove named entries; `_all` clears
estimates table NAMES     compact side-by-side comparison table
                          (options: se, t, p, star, stats(N r2 rmse F ...))
```

Aliased as `est`. Workspace gains a linked-list field `stored_est` (in `dataset.h`); freed by `ws_free`. Built on the existing `est_clone` infrastructure from the `hausman` work.

estout — publication-ready tables (new module `src/estout.c`):

```
estout [names] [using FILE] [, format(latex|markdown|md|plain)
                               se|t|p stars stats(...) nogaps
                               title("...") label(tab:foo)
                               keep(varlist) drop(varlist)
                               nomtitles mtitles("M1" "M2" ...)]
```

Default format is LaTeX (per the IMF / academic-economics convention). LaTeX output produces a tabular block with significance stars in the superscript style ($\hat{\alpha}^{***}$), variable names properly escaped, and a footnote legend. Stata-style `\_cons` for the constant term. Markdown and plain text variants also supported.

Verified end-to-end on a three-model comparison with the standard options: stars apply z-distribution thresholds (10/5/1%), SEs go in parentheses below each coef, stats footer rows handle `N/r2/r2_a/rmse/F/ df_r/df_m/sigma_u/sigma_e/rho`.

log audit and fix:

The `log` command captures output via a `printf`-macro override in `commands.c`. Problem: `display` lives in `interp.c` and was using bare `printf`, so its output went to `stdout` but not the log file. Fixed by making `g_logfp` extern and adding a manual tee from `do_display`. Round-trip verified: `log using file.log` now captures both command echoes (`. display ...`) AND their output (`_b[x] = 2`).

.dta round-trip audit:

Spot-checked save → use preservation: - Variable types: `□` - Variable formats (%8.2f, %td):
`□` - Variable labels (label variable x "GDP per capita"): `□` - Value labels (label define,
label values): `□` - Numeric values: `□` (down to floating-point precision) - String columns:
`□`

Not preserved in .dta: xtset panel/time variable assignment. Stata writes this as a dataset characteristic (`_dta[iis]`, `_dta[tis]`); readstat doesn't expose characteristic writing in its public API, and hand-writing the characteristic block in raw .dta format is a v1.1 project. Workaround for v1.0: re-run `xtset country year` after use, which is also the standard Stata-do-file convention.

Tier 2 — factor variables: `i.`, `c.`, `#`, `##`, `ib.` (NEW)

The Stata factor-variable grammar, now supported across every estimator that goes through `tsop_expand_varlist` (`regress`, `xtreg fe/re`, `logit`, `probit`, `predict`, `margins`). Lives in new module `src/factor.[ch]`.

Grammar:

<code>i.var</code>	dummies for each non-base level (base = smallest by default)
<code>ib<n>.var</code>	same, but base = level <code>n</code>
<code>c.var</code>	explicit continuous (display prefix; mostly for use in <code>#</code>)
<code>A#B</code>	interaction: cross product of A's and B's columns
<code>A##B</code>	main effects + interaction: equivalent to "A B A#B"

Examples (assume country has levels {1,2,3}, year has {2010,2011,2012}):

<code>i.country</code>	→ 2 cols: "2.country" "3.country"
<code>ib2.country</code>	→ 2 cols: "1.country" "3.country"
<code>i.country#c.gdp</code>	→ 2 cols: "2.country#c.gdp" "3.country#c.gdp"
<code>i.country#i.year</code>	→ 4 cols: "2.country#2011.year" "2.country#2012.year" "3.country#2011.year" "3.country#2012.year"
<code>i.country##c.gdp</code>	→ 5 cols total: 2.country, 3.country, gdp, 2.country#c.gdp, 3.country#c.gdp

Architecture:

- `factor.c` is a sibling to `tsop.c`: tokens are parsed into atoms (kind 'T' for `i.`, 'C' for `c.`, 'L' for coef-name "2.x", or bare variable in a `#` context). Atoms join into terms via `#`. `##` is unrolled to the power-set of single-term products (main effects + all interactions).
- `factor_is_factor_token` is the dispatch test; `factor_expand_token` produces a snapshot-able list of temp frame columns.
- Plumbed into `tsop_expand_varlist`: each token is tried as a TS-op first; if that returns "not mine," tried as a factor-var; if that returns "not mine," falls through to plain `varlist_expand`.

Predict/margins re-materialisation (the subtle bit):

When `predict` sees a coefficient named `2.country` or `2.country#c.gdp`, the column doesn't exist anymore — it was a temp dropped after the estimator finished. Solution: treat coef-name form `<level>.<varname>` as a new atom kind 'L' (level-pinned indicator). When `tsop_expand_varlist` is called with `"2.country"`, it dispatches to `factor`, which produces a single column with values `1{country==2}`. Same mechanism handles `2.country#c.gdp` (indicator × continuous).

Verified:

Spec	Hand-checkable result
regress y x i.country on y = $\alpha_c + x$	$\beta=1$, $_cons=\alpha_1$, 2.country= $\alpha_2-\alpha_1$, ...
regress y x ib2.country	$_cons=\alpha_2$, 1.country= $\alpha_1-\alpha_2$
regress y c.x#c.x on y = x^2	coef = 1.0 exactly
regress y i.country##c.x on y = $\alpha_c + \beta_c \cdot x$	reproduces all α and β exactly
regress growth L.growth i.cid (LSDV) on WEO	matches xtreg fe exactly

predict after logit y x i.g now works (was the open bug from the previous session). margins, dydx(*) after factor-var estimation works.

Restrictions in v1.0:

- i.var requires var to be numeric. Use encode for string ids.
- Composition with TS-ops (e.g., i.country#c.L.gdp) is not supported. Manually pre-compute the lag (gen L1_gdp = L.gdp) and use c.L1_gdp.
- Cell-count cap of 10000 columns per token (huge interactions blocked).
- Variable-name length limit of 32 chars means very long interaction names truncate. In practice not a problem for typical use.

Tier 1 polish — statistical functions, egen extensions, _b/_se, formats

A pass over the everyday-use surface in preparation for v1.0. An audit showed that tea's expression-language function library was already very comprehensive (math, trig, hyperbolic, strings, dates, distributions, glob+regex, value/variable labels, encode/decode/destring/tostring, recode, format). The gaps that mattered:

New statistical functions (in src/eval.c): - normalden(x) / normalden(x, σ) / normalden(x, μ , σ) — normal PDF - lnnormal(x) — log $\Phi(x)$, stable for large negative x via the asymptotic expansion (reuses tea_log_normal_cdf) - lnnormalden(x) — log $\phi(x)$ closed-form

Better regex (in src/eval.c): - regexs(n) — extract n-th submatch from the last regexm call. Uses a file-scope tea_regex_submatch[20] buffer. Index 0 is the whole match; 1..N are capture groups. - regexr(s, pat, rep) — replace first regex match with a string

Extended egen (in src/commands.c): - New aggregators: median, iqr, p25, p75, p50, pc (with option p(N) to specify percentile), rank, tag - Multi-argument row functions now work: rowtotal(v1 v2 v3), rowmean, rowmin, rowmax, rowsd, rowmiss, rownonmiss (the previous implementation was single-arg only) - tag(g1 g2) marks the first row of each group as 1, others 0 - rank(x) uses average-rank-for-ties

Format application (in src/commands.c::fmt_cell): Custom printf-style numeric formats set via format var %w.dF are now honored by list (previously only date formats like %td were).

Coefficient/SE macros after estimation (in src/regress.c + src/interp.c):

_b[name] and _se[name] are now accessible everywhere e(N) is. The estimators (regress, xtreg, logit, probit) store macros keyed by _b[varname] and _se[varname] (including bracketed names like _b[L.growth] for TS-op regressors). macro_expand recognises the pattern and substitutes outside double quotes; inside quotes the text stays literal. Standard Stata behaviour.

This unlocks idioms like:

```
quietly regress y x
display "t-stat = " _b[x] / _se[x]
gen yhat = _b[_cons] + _b[x]*x
```

Bug fix — quoted-string macro substitution (in src/interp.c): display "e(N) = " e(N) used to produce "3 = 3" because the macro expander substituted both occurrences of e(N), including the one inside the quoted string. Fixed by tracking in_dquote state and skipping e() / r() / _b[] / _se[] substitution inside strings. Stata's behaviour preserved: backtick-locals and \$globals are still substituted inside strings, since those are Stata's quote-aware substitution syntax.

margins — marginal effects after regress / logit / probit (NEW)

```
margins, dydx(*)           AME for all continuous regressors
margins, dydx(x1 x2)       AME for specified regressors
margins, dydx(*) atmeans    MEM (effects at sample means)
margins, dydx(x1)          AME for x1 only
```

After OLS, the AME of x_k is exactly β_k and the SE is exactly $SE(\beta_k)$ — the margins output reproduces the coefficient table. Useful as a sanity check.

After logit / probit, AMEs are the economically meaningful number: how the predicted *probability* changes with x , averaged across the sample. Coefficients themselves aren't directly interpretable (they're log-odds for logit, latent-index units for probit).

Math:

Logit: $m_k = (1/N) \sum_i \Lambda(X_i\beta) \cdot (1 - \Lambda(X_i\beta)) \cdot \beta_k$ Probit: $m_k = (1/N) \sum_i \varphi(X_i\beta) \cdot \beta_k$ OLS: $m_k = \beta_k$

Delta-method SE: $V(m_k) = G_k \cdot V(\beta) \cdot G_k'$, where G_k is the K-vector $\partial m_k / \partial \beta_j = (1/N) \sum_i [g'(\eta_i) X_{ij} \beta_k + g(\eta_i) 1\{j=k\}]$. Uses the full $V(\beta)$ matrix from the previous estimation (so robust/cluster SEs flow through automatically).

Verified: - Logit and probit on the same data produce nearly identical AMEs (a known result — they only differ at the tails, which contribute little to the average). In one test case $AME(\text{logit}, x1) = 0.12237$, $AME(\text{probit}, x1) = 0.12253$ — different in the fifth decimal. - AME for OLS reproduces the regress coefficient table exactly. - TS-op coefficient names (L.x etc.) work via the same snapshot-then-drop trick predict uses.

Deferred for v2: - margins x1 (predicted outcomes at varying x1 — useful for plots) - margins, at(x1=(0 1) x2=10) (predictions at specified values) - Discrete-change effects for indicator variables - Marginal effects after xtreg (would need within-effect-aware formulas)

predict extended to logit, probit, and xtreg (NEW)

The old predict worked only after regress, with options xb and residuals. Rewritten to dispatch on the previous estimator's cmd:

```
After regress:
  predict yhat           xb (default)
  predict r, residuals    y - Xβ
  predict s, stdp         standard error of the linear prediction

After logit / probit:
  predict p              predicted probability (default)
  predict idx, xb        linear index Xβ
```


After `xtreg (fe or re):`

<code>predict yhat</code>	<code>xb</code> (default; does NOT include <code>u_i</code>)
<code>predict u, u</code>	$\hat{\alpha}_i$ (panel effect)
<code>predict e, e</code>	idiosyncratic residual $y - X\beta - \hat{\alpha}_i$
<code>predict ue, ue</code>	$\hat{\alpha}_i + e$ ($= y - X\beta$)
<code>predict xbu, xbu</code>	$X\beta + \hat{\alpha}_i$ (in-sample prediction)

The dispatch lives in `predict_resolve_kind` and the per-row evaluation is one big switch. Sensible errors when options don't match the preceding command (e.g. `predict p, pr` after `regress` fails cleanly).

TS-op coefficient names: when `e->xnames[j]` is `L.growth` (TS-op form), the column doesn't exist in the frame anymore at predict time, but we can re-materialize it via the shared `tsop_expand_varlist`. A subtlety: we have to **snapshot** the materialized data into our own buffer, drop the temps, THEN add the new prediction variable — otherwise `tsop_drop_temps` would drop the just-added prediction column instead of the temp. This is what an earlier draft got wrong (a regression test would have shown a list-not-found error).

xtreg specifics: $\hat{\alpha}_i$ is computed in one pass over the in-sample rows (`e->used` tells us which) by streaming through the panel-sorted data and accumulating $\hat{y} - \bar{x}'\beta$ per panel. For predictions on rows that weren't in the estimation sample, $\hat{\alpha}$ is unknown so `u/e/ue/xbu` return missing.

Verified on the hand-checkable FE case (3 panels $\times$ 4 obs, perfect fit) where the math is exact: $\beta=1$, $\alpha=10/20/30$, $e=0$, $xbu=y$. Also tested on WEO AR(1) growth where `predict` reproduces the fit `yhat + r = growth` exactly.

logit and probit — binary-outcome MLE (NEW)

```
logit y x1 x2 ... [if] [in] [weight], [vce(robust|cluster v)]
probit y x1 x2 ... [if] [in] [weight], [vce(robust|cluster v)]
```

Implemented via a Newton-Raphson MLE driver in `src/mle.[ch]` with family-specific callbacks for the score, weight, and log-likelihood contributions. Same driver should extend to `poisson`, `cloglog`, etc. with no changes to the iteration logic — just new `MleFamily` structs.

Output matches Stata's layout: iteration log, LR χ^2 , pseudo R^2 , coefficient table with z-stats, optional cluster adjustment note.

Standard errors: - (default): classical $(X'WX)^{-1}$ (asymptotic inverse-Hessian) - `vce(robust)`: HC1 sandwich - `vce(cluster v)`: CR1 cluster-robust on score

Numerical care: - $\log(1+e^\eta)$ computed stably as $\max(\eta, 0) + \log(1+e^{-|\eta|})$ — no overflow. - Probit's ϕ/Φ uses asymptotic Mills bound $\lambda(\eta) \rightarrow -\eta$ when $\Phi(\eta)$ underflows. Per-obs Fisher weight clipped at $1e8$. - Starting values: OLS solve, which also detects rank deficiency (collinear regressors marked omitted, held at 0 through iteration). - Step halving (up to 8 halvings) when full Newton step decreases $\square$. - Perfect-separation detected when $\square \rightarrow 0$; warning printed.

Verified: - Intercept-only on 50/50 sample: $\beta_{\text{cons}} = 0$ exactly. SE = 0.2 for logit, 0.1253 for probit (matches closed forms). - Real-effect data: classical $\beta_{\text{logit}}/\beta_{\text{probit}} \approx 1.81$ ratio reproduced. - Newton converges in 3-5 iterations for well-conditioned data.

Postestimation: `test` and `lincom` work on logit/probit estimates. (`test` uses F with $df_r = N-K$; the asymptotic test would be χ^2 , but the difference is negligible for typical sample sizes.)

test syntax extended (NEW)

The test command now supports the full Stata-style hypothesis syntax:

```
test var                H0:  $\beta_{\text{var}} = 0$ 
test var = 0            same, explicit
test var = 0.5          H0:  $\beta_{\text{var}} = 0.5$  (numeric RHS)
test var1 = var2        H0:  $\beta_{\text{var1}} = \beta_{\text{var2}}$  (equality)
test var1 var2 ...      joint H0: all listed = 0
```

Internally constructs the restriction matrix R and value vector r , then computes the Wald $F = (R\beta - r)'(RVR')^{-1}(R\beta - r)/q$ against $F(q, df_r)$. Pre-existing test L.growth L2.growth (joint zero) is a special case and still works.

lincom accepts TS-op coefficient names (NEW)

The lincom parser now treats . as part of a coefficient name when followed by a name character, so lincom L.growth + L2.growth works correctly. Previously the parser stopped at the first . and tried to look up “L” as a coefficient.

xtreg, re — random-effects (GLS) estimator (NEW)

Implements panel RE via feasible GLS with quasi-demeaning:

```
xtreg y x1 x2 ... [if] [in], re [vce(robust|cluster v)]
```

Method: the model is $y_{it} = \alpha_i + x_{it}'\beta + \varepsilon_{it}$ with $\alpha_i \sim (0, \sigma_u^2)$ assumed independent of x_{it} . The estimator transforms each variable by subtracting θ_i times the panel mean, where:

$$\theta_i = 1 - \sigma_e / \sqrt{(T_i \cdot \sigma_u^2 + \sigma_e^2)}$$

Then runs OLS on $(y, X, 1-\theta_i)$ — note an explicit constant column, unlike FE. σ_e^2 comes from the within (FE) regression; σ_u^2 comes from the variance components computation ($\max(0, \text{var}(\hat{\alpha}_i) - \sigma_e^2/\bar{T})$). The transform interpolates between OLS ($\theta=0$) and FE ($\theta=1$).

Output matches Stata’s xtreg, re layout: - “Random-effects GLS regression” header - R-within, R-between, R-overall — all computed with $\beta_{\text{RE}} - \sigma_u, \sigma_e, \rho$ — variance components (shared with FE math) - θ — quasi-demeaning factor; min/avg/max for unbalanced panels - $_cons$ coefficient and SE (unlike FE) - Wald χ^2 for the slopes - $\text{corr}(u_i, X) = 0$ noted as the RE assumption

Verified on: - Degenerate hand-checkable case (perfect within fit $\rightarrow \theta=1 \rightarrow$ RE collapses to FE; $\beta_{\text{RE}} = \beta_{\text{FE}} = 1.0$ exactly; $_cons$ omitted since the $1-\theta$ column is identically zero). - WEO AR(1) growth: $\beta_{\text{RE},L} = 0.245$ sits between pooled OLS and FE $\beta = 0.220$. $\sigma_u/\sigma_e/\rho = 0.97/5.37/0.032$ — most variance within- panel. θ varies 0.19-0.39 across 210 countries.

v1 limitations: - Uses $\sigma_u^2 = \max(0, \sigma_{\text{BE}}^2 - \sigma_e^2/\bar{T})$ with $\bar{T} = N/n$ for unbalanced. Stata’s exact Swamy-Arora uses a slightly different $\bar{T}$ correction; for balanced panels the estimators agree.

hausman — FE vs RE specification test (NEW)

```
xtreg y x1 x2 ..., fe
xtreg y x1 x2 ..., re
hausman
```

Computes $H = (\beta_{FE} - \beta_{RE})' [V_{FE} - V_{RE}]^{-1} (\beta_{FE} - \beta_{RE})$, distributed $\chi^2(K_{common})$ under H_0 : $cov(\alpha_i, x_{it}) = 0$. Rejection means RE is inconsistent; use FE.

Tea saves the last xtreg fe and last xtreg re into dedicated workspace slots automatically, so the user just types hausman with no arguments after running both. Order doesn't matter.

Implementation notes: - Only compares slope coefficients present (and not omitted) in both estimates. `_cons` is dropped since FE doesn't have one. - The difference matrix $V_{FE} - V_{RE}$ should be PSD under H_0 , but sampling noise (especially with robust/cluster SEs) can make it not strictly PD. Cholesky inverse is tried first; if it fails, falls back to an SVD pseudoinverse. When the resulting statistic comes out negative, we report $|H|$ with a "sign flipped" warning. - Verified on WEO AR(1) growth: $\chi^2(1) = 189$, $p < 0.0001 \rightarrow$ strongly rejects RE (matches the FE-reported $corr(u_i, Xb) = 0.97$ warning sign). On a constructed case where $\alpha_i \perp x$ by construction, $\chi^2 \approx 0$ and the test correctly fails to reject.

xtreg, fe — fixed-effects (within) estimator (NEW)

Implements panel FE OLS via within transformation:

```
xtreg y x1 x2 ... [if] [in], fe [vce(robust|cluster v)]
```

Reports R-within, R-between, R-overall, σ_u , σ_e , ρ , $corr(u_i, X\beta)$, and the F-test that all $u_i = 0$. Verified on a hand-checkable case (tests/regression/29_xtreg_fe.do): 3 panels $\times$ 4 obs, $y = \alpha_i + x$ with $\alpha \in \{10, 20, 30\}$ gives $\beta_{FE} = 1.0$ exactly, $R_{within} = 1.0$, $\sigma_u = 10$ exactly. Verified against the WEO AR(1) growth dynamic showing internally consistent results (β from xtreg D.growth L.growth, fe = $-(1 - \beta$ from xtreg growth L.growth, fe)).

Standard error options: | , fe | classical: $\sigma^2 \cdot (X_w' X_w)^{-1}$ with $df = N - n_{groups} - K$
 | | , fe vce(robust) | HC1 sandwich on within-transformed data | | , fe vce(cluster v)
 | CR1 cluster-robust on within-transformed data |

v1 limitations (deferred to v2): - `_cons` is not displayed. Its point estimate (mean of α_i) is computable but the strictly correct SE requires LSDV-equivalent variance propagation we haven't implemented. Users wanting inference on `_cons` can run `regress y x i.panel` for now. - Stata defaults `vce(robust)` to cluster-by-panel for xtreg fe; tea's `vce(robust)` is HC1. Use explicit `vce(cluster panelvar)` for cluster-by-panel. - xtreg, be (between effects) is not yet implemented — returns a clean error suggesting , fe or , re. (, re ships in this version — see above.) - Weights work the same way as in regress (build_design handles them), but the FE-specific weight semantics have not been carefully verified yet.

TS-op handling unified across all varlist-accepting commands (NEW)

Time-series operator support is now centralized in `src/tsop.[ch]`. The parser handles every Stata form once (`L.x`, `L2.x`, `L(1/2).x`, `L.(x y)`, `L2.(x y)`, `L(1/2).(x y)`, step form `L(1(2)9).x`), and every command that accepts a varlist routes through `tsop_expand_varlist`. Concretely:

Command	TS ops	Mechanism
regress	□	tsop_expand_varlist (depvar + xspec)
summarize	□	tsop_expand_varlist
list	□	tsop_expand_varlist
tabulate	□	tsop_expand_varlist
tabstat	□	tsop_expand_varlist
collapse	□	tsop_expand_varlist (paren-aware tokenizer)

regress lost ~400 lines of duplicated parser code in this refactor (from 1198 down to 796 lines). Future commands that accept varlists — xtreg, ivregress, correlate, etc. — get TS-op support for free by calling `tsop_expand_varlist`. Test 28 exercises the universal coverage; tests 21-27 cover individual command surfaces.

Side fix discovered: keep if / drop if used to call `frame_unsort` which over-eagerly cleared `ts_panel / ts_time`. Row deletion doesn't invalidate panel structure (the columns are still there and still sorted; gaps are fine for tea's gap-aware `tsidx` lookup), so the ts state now survives those operations.

xtdescribe / xtdes (NEW)

Added `xtdescribe` (with `xtdes` abbreviation) to summarize the panel structure declared by `xtset`. Shows number of panels, time range, delta, span, min/p25/p50/p75/max distribution of obs per panel, and balanced/unbalanced determination. Test 27 exercises both balanced and unbalanced cases.

TS operators in summarize and list (FIXED)

`summarize L.x` and `summarize D.x` failed with “variable not found” because those commands called `varlist_expand` directly and didn't recognize TS-op tokens. The earlier TS-op fix only touched `regress`.

Fixed by adding a shared `tsop` module (`src/tsop.[ch]`). Both `summarize` and `list` now route their varlist through `tsop_expand_varlist`, which materializes TS-op tokens as temporary frame columns (with canonical display names like `L.x`, `D2.y`) and appends them to the frame for the duration of the command, then drops them via `tsop_drop_temps` before returning. Test 26 covers this.

D-operator iterated differencing (FIXED — silent math bug)

`D2.x`, `D3.x`, etc. all returned the same as `D.x` because the eval code hardcoded `back = -1` for any D-kind operator. This was a **silent** numerical bug: regressions including `D2.x` would compute correctly for `D.x` but the higher-order differences were all collinear with `D.x` and got “omitted” — looking like a model specification issue rather than a math error.

Stata's convention is that `D#.x` is the iterated #-th difference:

- $D.x = x[t] - x[t-1]$
- $D2.x = D(D.x) = x[t] - 2 \cdot x[t-1] + x[t-2]$
- $D3.x = x[t] - 3 \cdot x[t-1] + 3 \cdot x[t-2] - x[t-3]$
- $D^k x[t] = \sum_{j=0..k} (-1)^j \cdot C(k,j) \cdot x[t-j]$

Fixed in `src/eval.c` using a binomial-recurrence loop. The `S` operator (separate semantics: simple gap-aware seasonal difference, $S#.x = x[t] - x[t-#]$) is unchanged. Test 25 verifies on $x = t^2$ where `D.x` is linear, `D2.x` is constant 2, `D3.x` is constant 0.

Action item for users: any analysis using `D2.x`, `D3.x`, or higher in past sessions silently produced wrong values. Re-run any such code.

Operator-list TS syntax (FIXED)

Stata's compact form `regress y L(1/2).x` (expanding to `L1.x L2.x`) was rejected with “independent variables not found”. Tea now handles the full set of Stata TS-op forms in `regress` varlists:

Syntax	Expands to
<code>L.x</code>	<code>L.x</code>
<code>L2.x</code>	<code>L2.x</code>
<code>L0.x</code>	<code>x</code> (no shift)
<code>L(1/3).x</code>	<code>L.x L2.x L3.x</code> (range)
<code>L(1 3).x</code>	<code>L.x L3.x</code> (explicit list)
<code>L(1(2)9).x</code>	<code>L.x L3.x L5.x L7.x L9.x</code> (step form)
<code>L.(x y)</code>	<code>L.x L.y</code>
<code>L2.(x y)</code>	<code>L2.x L2.y</code>
<code>L(1/2).(x y)</code>	<code>L.x L.y L2.x L2.y</code> (cross product)

Implemented in `src/regress.c` via the unified `try_tsop_form` parser plus `parse_numlist` (range / explicit list / step) and `find_matching_paren` for nested parens. The regressor tokenizer is paren-aware so spaces inside (...) don't break tokenization. Tests 21, 23, and 24 cover the forms.

Multi-digit time-series operators read wrong lag (FIXED)

`L2.x` was actually computing the 12th lag instead of the 2nd; `L3.x` the 13th, `F2.x` the 12th lead, `S3.x` a 13-period seasonal difference. Affected expressions, `regress`, and any other consumer.

The bug was in `try_tsop` in `src/parse.c`: the lag-magnitude accumulator was initialized to 1 instead of 0, so reading “L2” gave `num = 1*10 + 2 = 12`. Lag-1 (`L.x / L1.x / no digits`) accidentally worked because `num` stayed at 1 either way.

Fixed by initializing `num = 0`. The existing `if (!have) num = 1`; fallback still gives bare `L.` the right value of 1. Test 22 pins down correct multi-digit operator behavior.

Time-series operators rejected in regress varlist (FIXED)

`regress growth L.growth if country_id == "USA"` failed with “regress: independent variables not found” because the regress code called `varlist_expand` directly on each regressor token; that helper only knew about plain names, wildcards, ranges, and `_all` — not the `L.x / F.x / D.x / S.x` syntax.

Fixed by refactoring `build_design` in `src/regress.c` to detect TS-op tokens and route them through the existing expression evaluator (`expr_parse` → `N_TSOP`). Plain names still go through `varlist_expand`. Works for both the dependent variable and the regressors (`regress D.growth L.growth` is supported). Test 21 covers `regress` with TS operators.

capture and _rc (FIXED in v1.4.0)

Fixed: `capture` now suppresses all output of the captured command (stdout and stderr, at the file-descriptor level) and `_rc` returns the captured return code — 0 on success, the error code otherwise — exactly as in Stata. One small remaining deviation: a successful command that is NOT under capture leaves `_rc` unchanged rather than resetting it to 0; branch on `_rc` only immediately after a capture.

test 08 captures fewer lines than expected

The reshape-error regression test passes, but if you read the expected output you'll notice some error messages print to stderr while others to stdout, and the line ordering across

the merged stream depends on line-buffering. This is OK as of v0.5.20 because we set the stdout buffer to line-buffered in `main.c`, but the harness assumes the ordering is stable — if a future change reorders error/output emission between commands, the test may need updating.

Storage types are display-only

`gen byte x = 1` accepts the byte qualifier but stores the value as double internally; `gen strl0 country = "USA"` accepts `strl0` but doesn't track width. This is a deliberate design decision documented in `COMPATIBILITY.md` — tea is double + string internally regardless of declared type, and the `.dta` writer auto-compresses on save. Not a bug.

Numeric function family coverage

Many Stata numeric functions are not yet implemented:

- Distribution functions: `normal`, `normalden`, `invnormal`, `chi2`, `chi2tail`, `invchi2`, `t`, `ttail`, `invttail`, `F`, `Ftail`, `binomial`, `binomialp`, `poisson`, `poissonp`.
- Matrix functions (we don't have a matrix type yet).
- Random-number generators: `runiform`, `rnormal`, etc. (Tied to `set seed`.)

The existing function set covers basic arithmetic, transcendentals, date/time, string manipulation, and the common aggregator functions used in `egen`.

v0.6 .dta polish (this milestone)

- **Value labels** — `label define` / `label values` sets are now written into `.dta` and read back via `use`. Round-trips cleanly (test 16).
- **Dataset label** — `label data "text"` now persists. Visible in `describe`, round-trips through `save/use` (test 17).
- **save FILE.dta, version(NUM)** — emit any DTA format 104-119 (test 18). Defaults to 118 (Stata 14).
- **Workspace label-set cleanup** — `clear` and `use`, `clear` now also drop the workspace's value-label sets, matching Stata semantics.

Still deferred from v0.6

- **Notes** — Stata's note: text on a dataset or variable. Not yet read or written.
- **strL strings > 2045 bytes** — currently strings longer than the `str#` limit are truncated at 2045 on write. Real-world risk is essentially zero for econ data; will add when needed.
- **.dta fweight metadata writes are silently dropped** — `readstat 1.1.9` accepts `readstat_writer_set_fweight_variable` but does not serialize the resulting Stata `_fweight` characteristic. Tea's read-side captures the metadata correctly when present (so real Stata-saved files round-trip through tea correctly *if* the user doesn't save through tea), and the write-side wiring is in place for when `readstat` fixes this. Track upstream.

reported by user during real-data testing

- Bug 1 — `_N` in display context returned 1 → **FIXED in v0.5.20**
- Bug 2 — `tabstat columns()` ignored in `by`-mode → **FIXED in v0.5.19**
- Bug 3 — `tabstat` with `if` + `by` showed empty `by`-groups → **FIXED in v0.5.18**

- Bug 4 — list * segfaulted on 147-var dataset → **FIXED in v0.5.14**
- Bug 5 — quoted filename in import not opened → **FIXED in v0.5.12**

reported by Claude during synthetic testing

- Bug 6 — commas in string literals broke parser → **FIXED in v0.5.20**
- Bug 7 — \$1 in strings expanded as undefined macro → **FIXED in v0.5.20**
- Bug 8 — egen silently skipped strings → **FIXED in v0.5.20**

reported by Mico during WB fertility replication (v1.6.x)

- Bug 9 — import excel, firstrow header naming didn't match Stata: invalid chars were underscored (Country Code → Country_Code) instead of removed (CountryCode), and digit-leading headers (1960) got a v prefix instead of Stata's Excel column-letter fallback (F...BQ). Do-files written against Stata failed with variable not found. → **FIXED**: excel/ods import now uses Stata's exact rule; duplicate headers fall back to the column letter.
- Bug 10 — foreach v of varlist BAD was a **silent no-op**: an unknown variable made varlist_expand return -1, which the loop treated as an empty list. The body ran zero times with no error, and scripts failed much later with a misleading message. → **FIXED**: hard error, rc=111. The same -1-as-empty flaw was guarded at every remaining silent site: by BAD: (ran ungrouped!), egen ..., by(BAD) (ungrouped), collapse, by(BAD) (would collapse the whole dataset to one row), format % BAD (silent no-op). All now rc=111.
- Bug 11 — local ++yr parsed ++yr as the macro *name*, silently defining a macro literally called ++yr while yr never changed — loop counters froze and rename loops died with “already exists” on the second iteration. → **FIXED**: local ++x / local --x now increment/decrement per Stata; ++ on an undefined or non-numeric macro errors (rc=198) instead of silently minting a new macro.